



# MATHEMATICS

## Preliminary Course Assessment Task 1

March 8, 2013

### General instructions

- Working time – 50 minutes.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the writing paper used in the correct order within this paper and hand to examination supervisors.
- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NAME: ..... # SHEETS USED: .....

Class (please ✓)

☐ 11M2A – Mr Lin

Marker's use only.

| QUESTION(S) | 1              | 2               | 3               | 4              | 5              | 6               | Total           | % |
|-------------|----------------|-----------------|-----------------|----------------|----------------|-----------------|-----------------|---|
| MARKS       | $\overline{7}$ | $\overline{20}$ | $\overline{15}$ | $\overline{7}$ | $\overline{8}$ | $\overline{13}$ | $\overline{70}$ |   |

**Question 1 (Commence a new page)**

- a) (i) Evaluate  $\sqrt[3]{23.6}$  correct to one decimal place. 1
- (ii) Express  $3 \times 10^{-5}$  as a fraction. 1
- b) Express 346.705529 correct to
- (i) 3 decimal places. 1
- (ii) 3 significant figures. 1
- c) Voyager Spaceship has travelled 75 960 000km. 1  
Express this distance in scientific notation.
- d) Express the following as common fractions in their simplest form.  
Show all necessary working.
- (i)  $0.\dot{6}$  1
- (ii)  $0.58\dot{3}$  1

**Question 2 (Commence a new page)**

- a) Expand and simplify
- (i)  $4(2t - 1) + t(3t + 2) - t(3t^2 - 2)$  2
- (ii)  $(7r + 2)(7r - 2)$  1
- (iii)  $(3x + 4)^2 - (2x - 5)^2$  2

b) Factorise completely

(i)  $4a + 4b - ax - bx$  **2**

(ii)  $72 - 2x^2$  **2**

(iii)  $5k^3 + 40$  **1**

(iv)  $2x^2 + 11x + 12$  **2**

c) Simplify

(i)  $\frac{a}{a-b} + \frac{b}{b-a}$  **2**

(ii)  $\frac{x^3 - 1}{x^2 - 1} \div \frac{3x^2 + 3x + 3}{x^2 - 4x - 5}$  **3**

d) Simplify  $\frac{1}{x^2 - 4} + \frac{1}{x + 2} - \frac{5}{3x - 6}$  **3**

**Question 3 (Commence a new page)**

a) Simplify:

(i)  $\sqrt{18} - \sqrt{50} - \sqrt{72}$  **2**

(ii)  $(\sqrt{3} - 2)(4\sqrt{3} - 6)$  **2**

(iii)  $9\sqrt{6} \times 7\sqrt{6}$  **1**

(v)  $(\sqrt{5} - 3\sqrt{2})^2$  **2**

b) Express as a surd with a rational denominator.

(i)  $\frac{\sqrt{5} - 2}{3\sqrt{5}}$  **2**

(ii)  $\frac{2\sqrt{5} - \sqrt{2}}{3\sqrt{5} + 2\sqrt{2}}$  **3**

c) Given  $x = \sqrt{5} + 2$ , find  $b$  if  $x + \frac{1}{x} = 2\sqrt{b}$  **3**

**Question 4 (Commence a new page)**

a) Solve:

(i)  $\frac{x+6}{5} = 3$  **1**

(ii)  $\frac{x+5}{2} - \frac{x-1}{3} = 1$  **2**

(iii)  $2(3x+7) = 6 - (x-1)$  **2**

b) The Fahrenheit temperature Suzie need to cook at is  $383^{\circ}\text{F}$ . **2**  
However she has an oven with Celsius range only.

Using the formula  $F = \frac{9C}{5} + 32$  (where  $F$  = degrees Fahrenheit and  $C$  = degrees Celsius), Suzie can convert to Celsius degrees.

What temperature in Celsius will she cook at?

**Question 5 (Commence a new page)**

a) Solve the following for x

(i)  $|x-3| = 5$  **2**

(ii)  $|2x+6| = 3x-1$  **3**

b) Solve the following equations simultaneously: **3**

$2x + 3y + 1 = 0$  and  $x + 2y = 4$

**Question 6 (Commence a new page)**

- a) Solve
- (i)  $x^2 - 5x - 6 = 0$  **1**
- (ii)  $3x^2 - 14x + 8 = 0$  **2**
- (iii)  $3x^2 - 4x - 1 = 0$  (leave answers in exact surd form) **2**
- b) Eight more than twice a number is five less than the number itself.
- (i) Write an algebraic expression for this statement. **1**
- (ii) What is the number that satisfies this statement? **1**
- c) Solve  $3 \leq 2(6 + x)$  and graph its solution on a number line. **3**
- d) Solve  $x^2 - x - 30 < 0$  **2**
- e) Solve  $|4x + 10| > 13$  **2**

**End of examination**

**Question 1**

- (a) (i) 2.9 1
- (ii)  $\frac{3}{100000}$  1
- (b) (i) 346.706 1
- (ii) 347 1
- (c)  $7.596 \times 10^7$  1
- (d) (i)  $\frac{6}{9}$  1
- (ii)  $\frac{7}{12}$  1

**Question 2**

- (a) (i)  $8t - 4 + 3t^2 + 2t - 3t^3 + 2t$  1 (correct expansion)
- $= -3t^3 + 3t^2 + 12t - 4$  1 (correct answer)
- (ii)  $49r^2 - 4$  1
- (iii)  $9x^2 + 24x + 16 - 4x^2 + 20x - 25$  1 (correct expansion)
- $= 5x^2 + 44x - 9$  1 (correct answer)
- (b) (i)  $4(a + b) - x(a + b)$  1 (attempt at using 'grouping in pairs' method)
- $= (a + b)(4 - x)$  1
- (ii)  $2(36 - x^2)$  1
- $2(6 + x)(6 - x)$  1
- (iii)  $5(k + 2)(k^2 - 2k + 4)$  1
- (iv)  $(2x + 3)(x + 4)$  1
- (c) (i)  $\frac{a-b}{a-b}$  1
- $= 1$  1
- 
- (ii)  $\frac{(x-1)(x^2+x+1)}{(x+1)(x-1)} \times \frac{(x-5)(x+1)}{3(x^2+x+1)}$  2 (-1 for each mistake)
- $= \frac{x-5}{3}$  1 (correct answer)
- (d)  $\frac{3+3(x-2)-5(x+1)}{3(x+2)(x-2)}$  1
- $\frac{3+3x-6-5x-5}{3(x+2)(x-2)}$  1
- $\frac{-2x-8}{3(x+2)(x-2)}$  1

**Question 3**

- |         |                                                                          |                         |
|---------|--------------------------------------------------------------------------|-------------------------|
| (a) (i) | $3\sqrt{2} - 5\sqrt{2} - 6\sqrt{2}$                                      | 1                       |
|         | $= -8\sqrt{2}$                                                           | 1                       |
| (ii)    | $12 - 6\sqrt{3} - 8\sqrt{3} + 12$                                        | 1                       |
|         | $= 24 - 14\sqrt{3}$                                                      | 1                       |
| (iii)   | 378                                                                      | 1                       |
| (iv)    | $5 - 6\sqrt{10} + 18$                                                    | 1                       |
|         | $= 23 - 6\sqrt{10}$                                                      | 1                       |
| (b) (i) | $\frac{5-2\sqrt{5}}{15}$                                                 | 2                       |
|         |                                                                          | (-1 if not simplified)  |
| (ii)    | $\frac{30-4\sqrt{10}-3\sqrt{10}+4}{45-8}$                                | 2 (-1 for each mistake) |
|         | $= \frac{34-7\sqrt{10}}{37}$                                             | 1                       |
| (c)     | $\frac{5+4\sqrt{5}+4+1}{\sqrt{5}+2}$                                     | 1                       |
|         | $= \frac{10+4\sqrt{5}}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2}$ | 1                       |
|         | $= 2\sqrt{5}$                                                            |                         |
|         | $\therefore b = 5$                                                       | 1                       |

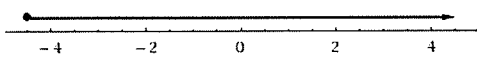
**Question 4**

- |         |                        |   |
|---------|------------------------|---|
| (a) (i) | $x = 9$                | 1 |
| (ii)    | $3x + 15 - 2x + 2 = 6$ | 1 |
|         | $x = -11$              | 1 |
| (iii)   | $6x + 14 = 6 - x + 1$  | 1 |
|         | $x = -1$               | 1 |
| (b)     | $195^{\circ}\text{C}$  | 3 |

**Question 5**

- (a) (i)  $x - 3 = 5$   $-(x - 3) = 5$   
 $x = 8$   $-x + 3 = 5$   
 $x = -2$  **2**
- (ii)  $2x + 6 = 3x - 1$   $-2x - 6 = 3x - 1$   
 $x = 7$   $5x = -5$   
 $x = -1$  **3**
- (b)  $y = 9, x = -14$  **3**

**Question 6**

- (a) (i)  $(x + 1)(x - 6)$  **1**
- (ii)  $(3x - 2)(x - 4)$  **2**
- (iii)  $x = \frac{4 \pm \sqrt{28}}{6}$   
 $x = \frac{2 \pm \sqrt{7}}{3}$  **2**
- (b) (i)  $2x + 8 = x - 5$  **1**
- (ii)  $x = -13$  **1**
- (c)  $3 \leq 12 + 2x$  **1**  
 $-2x \leq 9$  **1**  
 $x \geq -\frac{9}{2}$  **1**
- 
- (d)  $(x - 6)(x + 5) < 0$   
 $-5 < x < 6$  **2**
- (e)  $4x + 10 > 13$   $-(4x + 10) > 13$   
 $x > 0.75$   $x < -5.75$  **2**